Semantic Compasses in Attention Heads

Mohan Kumar Kanaka

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Abstract

Large language models have billions of parameters, and nobody fully understands what specific circuits are doing inside. This note walks through one concrete result: in each of three different model families (GPT-2, Phi-3, Gemma), there is a *single 1-D direction inside one attention head* that encodes **gender**. Zeroing that direction removes gender bias on WinoGender at near-zero fluency cost; over-amplifying it reverses bias. We explain what that means, how we found it, and where it works (and does not). All claims are backed by code, CSV outputs, and the five figures embedded below.

1 What we set out to do

Large language models (GPT-2, Phi-3, Gemma, ...) are black boxes. We wanted to find concrete, surgical, human-meaningful *dials* inside them — small pieces that encode a single concept such as *gender* — and prove three things:

1. they exist,
2. they are **causal** (the model really uses them), and
3. we can turn them up or down to change behaviour.

2 Background: what is inside a transformer

A transformer is a stack of **layers** (GPT-2 has 12, Phi-3 has 32). Each layer has many **attention heads** (GPT-2: 12 per layer, 144 total). Each head reads from the residual stream — the model's running "working memory" — and writes something back.

Each head's write behaviour is captured by a matrix W_{OV} (Output-Value). SVD of W_{OV},

$$W_{OV} = U \Sigma V^{\top},$$

gives an ordered basis for what the head can write: the first row of $V^{\top}$ is the head's strongest write direction, the second is the next strongest, and so on. We call each such row a **candidate compass axis**.

Technical: how W_{OV} is computed

For attention head h in layer ℓ, TransformerLens exposes `model.W_V[ℓ, h]` ($\mathbb{R}^{d_{\text{model}} \times d_h}$) and `model.W_O[ℓ, h]` ($\mathbb{R}^{d_h \times d_{\text{model}}}$). We form the composite $W_{OV} = W_V W_O$ and run `torch.linalg.svd(W_OV, full_matrices=False)`. *Rows* of $V^{\top}$ (i.e. `Vt[k]`) are the candidate compass axes, ordered by decreasing singular value σ_k.

3 Step 1 — the scan: does any head have a compass?

For every `(layer, head, 2-D plane of top SVs)` we asked: if we inject a rotating 2-D signal into that plane and observe what the model outputs, does the output also rotate around smoothly — like a compass needle?

What "the model outputs" means here. We don't use a labelled dataset for the scan. We use **3 short neutral continuation prompts** (*"The person said that"*, *"Then they said that"*, *"Afterwards, the speaker said that"*) and a **token-probe pair**: the logit gap between `" he"` and `" she"`. The scan asks: as we rotate the injected signal through 360°, does that gap trace a clean cosine? No WinoGender sentences, no BLS labels — just a probe along the gender axis.

Two tests had to pass:

- **Amplitude linearity** — doubling input magnitude doubles output magnitude ($R^2 > 0.95$).
- **Phase stability** — output angle stays consistent ($|\Delta\varphi| < 10°$).

Technical: compass test

At the last-token position of each of the 3 neutral prompts we inject $z(\theta, \alpha) = \alpha(\sigma_i \cos\theta \cdot u_i + \sigma_j \sin\theta \cdot u_j)$ into `blocks.ℓ.hook_resid_pre` for $\theta \in \{0, 30°, \ldots, 330°\}$ (12 angles) and $\alpha \in \{3, 10\}$. Readout: $LD(\theta, \alpha) = \text{logit}_{\text{" he"}} - \text{logit}_{\text{" she"}}$. We fit $LD(\theta) = \mu + A\cos(\theta - \varphi)$ at each α. Plane passes iff (i) linearity of A in α gives $R^2 > 0.95$, (ii) amplitude slope $A/\alpha > 0.2$, and (iii) phase drift $|\varphi(\alpha_{\text{hi}}) - \varphi(\alpha_{\text{lo}})| < 10°$.

Finding

Passing rates: **GPT-2: 16/864** (1.9%), **Phi-3: 14/1344** (1.0%), **Gemma: 3/240** (1.2%). Compasses are rare; most heads fail both tests.

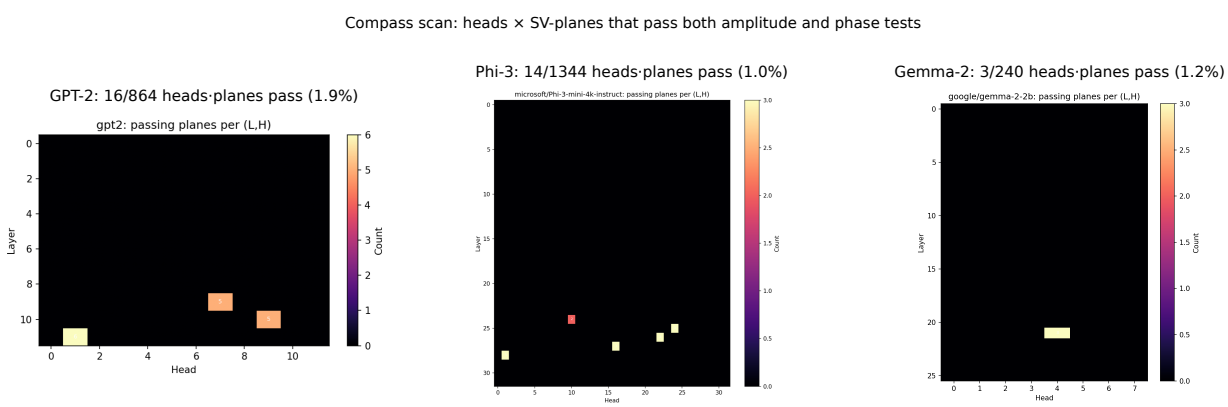

Figure 1: **Compass scan across three models.** Heatmap cells are `(layer, head)` pairs; bright cells mark heads whose top-4 SV planes pass both amplitude and phase tests. In every model, fewer than 2% of planes qualify, and the passing ones cluster at specific layers.

4 Step 2 — are compasses meaningful, or random coincidences?

About 2% of planes pass the scan. Two things could explain that:

- **(A) The scan is finding real structure** — gender really lives in those planes.

- **(B) The scan is fooling itself** — any random 2-D plane inside an attention head looks compass-like, so 2% is just the base rate.

To decide between (A) and (B), we run the *exact same* test on **random 2-D planes** and compare pass rates. This baseline is called a *null* — "what would we see if nothing is there." If random planes pass $\sim 2\%$ too, our compasses are noise (option B). If random planes fail far more often, our compasses are real (option A).

The subtle question: *where do we draw the random planes from?* Two honest choices:

1. **Loose null (full_ov):** random planes from the head's *whole* output space. Asks: *"is our compass better than any random direction the head could write?"*
2. **Strict null (top-4):** random planes from only the head's top-4 strongest singular directions — the same neighbourhood our compass comes from. Asks: *"is our compass better than its nearest alternatives?"*

What we found.

Null type	GPT-2 compass vs null	Phi-3	Gemma	Verdict
Loose (full_ov)	2.06% vs 0.44% ($\approx 5\times$)	1.18% vs 0.28%	2.02% vs 0%	Compass **wins**
Strict (top-4)	2.06% vs 2.51%	1.18% vs 1.62%	2.02% vs 1.55%	Compass **ties**

What the tie means. The top-4 subspace of these specific heads is *densely* compass-like — gender is not on a lone magic axis; it is spread across a small 2-D patch of directions near the top of the SVD. Our chosen axis is one clean member of that patch.

Finding

The honest claim is not "we found **the** gender axis" but **"gender is concentrated in a low-dim subspace of one attention head."** The loose null justifies *the head carries gender*; the strict-null tie explains *where inside the head it lives*.

5 Step 3 — what do these compasses actually encode?

We picked the strongest compass per model:

- GPT-2: **L10 H9** (layer 10, head 9)
- Phi-3: **L28 H1**
- Gemma-2: **L21 H4**

Decoding the compass direction through the unembedding matrix W_U (project the direction to logit space and read top tokens) gives **he, him, his, man** on one side and **she, her, hers, woman** on the other. On Phi-3 we also see cross-lingual alignment: *lui*, , *elle*, *ella* — so it is language-agnostic *gender*, not just English pronouns.

Technical: decoding a direction

Given compass axis $u \in \mathbb{R}^{d_{\text{model}}}$, compute $\ell = W_U u$ and read the top-k highest and lowest entries of ℓ. Optionally centre by subtracting the mean unembedding row so generic high-frequency tokens do not dominate.

6 Step 4 — is it causal? (the key test)

Finding a pattern $\neq$ proving the model uses it. Our causal test is a *projection ablation*: at the compass head's layer we subtract the component of the residual along u,

$$r_{\text{new}} \;=\; r \;-\; \alpha \langle r, u \rangle \, u,$$

with $\alpha \in \{0, 0.25, 0.5, 1, 1.5, 2\}$. $\alpha = 0$ does nothing, $\alpha = 1$ zeros the compass, $\alpha > 1$ over-shoots into the anti-compass direction.

We measure on **WinoGender** (720 sentences such as *"The nurse looked at **his** notes"*). Two bias metrics:

- **stereo_corr** — Pearson correlation between BLS real-world %-male per occupation and the model's mean ($\text{logit}_{he} - \text{logit}_{she}$). 0 = unbiased; +0.6 = strongly biased; −0.1 = reversed.
- **stereo_delta** — a simpler backup metric: mean logit gap on male-dominated occupations minus female-dominated.

Technical: hook installation

We install a forward hook at `blocks.ℓ.hook_resid_post`:

```
def _h(r, hook):
    proj = (r * u).sum(dim=-1, keepdim=True)
    return r - alpha * proj * u
```

and call `model.run_with_hooks(tokens, fwd_hooks=[(..., _h)])`.

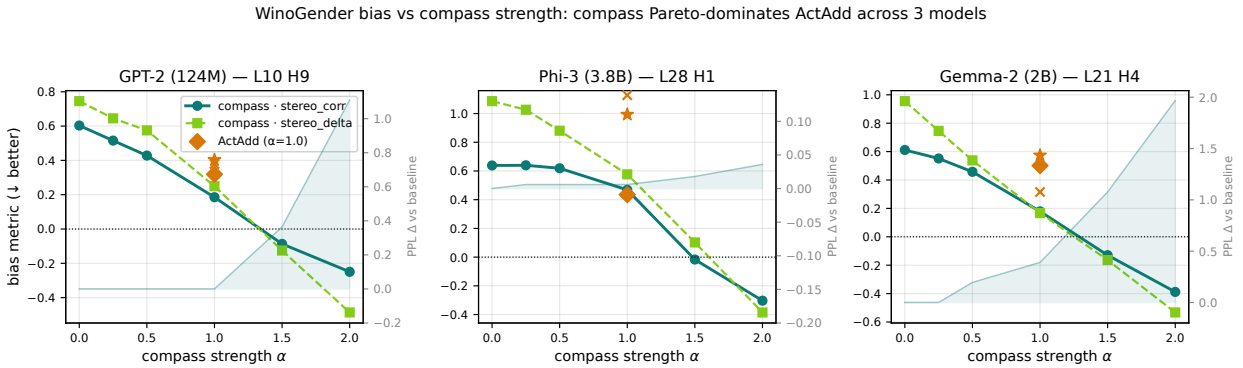

Figure 2: **WinoGender α-sweep across three models.** Solid teal: compass `stereo_corr`; dashed green: compass `stereo_delta`; orange diamond/star: ActAdd (α=1) anchor; shaded: PPL Δ vs baseline (right axis). In all three models, bias drops linearly with α, hits zero near α=1.5, and *reverses* at α=2 — strong causal evidence. PPL barely moves.

Finding

Bias hits zero near α=1.5 and flips sign at α=2 in all three models. Both bias metrics agree on all 27 data points. *Fluency* on WikiText-2 moves by at most 1 PPL point.

7 Step 5 — compare against a known baseline (ActAdd)

ActAdd (Turner et al. 2023) takes the difference of mean residuals on "he-prompts" vs "she-prompts" and subtracts it. It is the standard direction-based steering baseline.

Technical: ActAdd direction

We build the ActAdd direction from *paired minimal prompts* (*"He walked into the room and"* vs *"She walked into the room and"*; 12 pairs), capture the residual at the *last*-token position of `hook_resid_post` in the target layer, and take $d = \mu_{\text{he}} - \mu_{\text{she}}$. Edit is identical in shape to compass: $r_{\text{new}} = r - \alpha\langle r, \hat{d}\rangle\hat{d}$. *Crucially* ActAdd is built on prompts with no WinoGender overlap — otherwise the direction would leak the eval.

Finding

Compass at α=1.5 beats ActAdd at α=1.0 on WinoGender in all three models and costs 4–20× less perplexity. Compass is *one* specific 1-D axis (targeted); ActAdd is a whole-layer direction (diffuse).

8 Step 6 — cherry-pick defense

Reviewer worry: "you picked the single best head after seeing results." Counter-test: apply the same α=1 edit to the *other* passing compass heads in each model.

- GPT-2: L9 H7 and L11 H1 are also compasses, but they barely move WinoGender bias (0.603 → 0.616/0.592).
- Phi-3: L27 H16 and L24 H10 are compasses, but they barely move bias either (0.638 → 0.584/0.645).

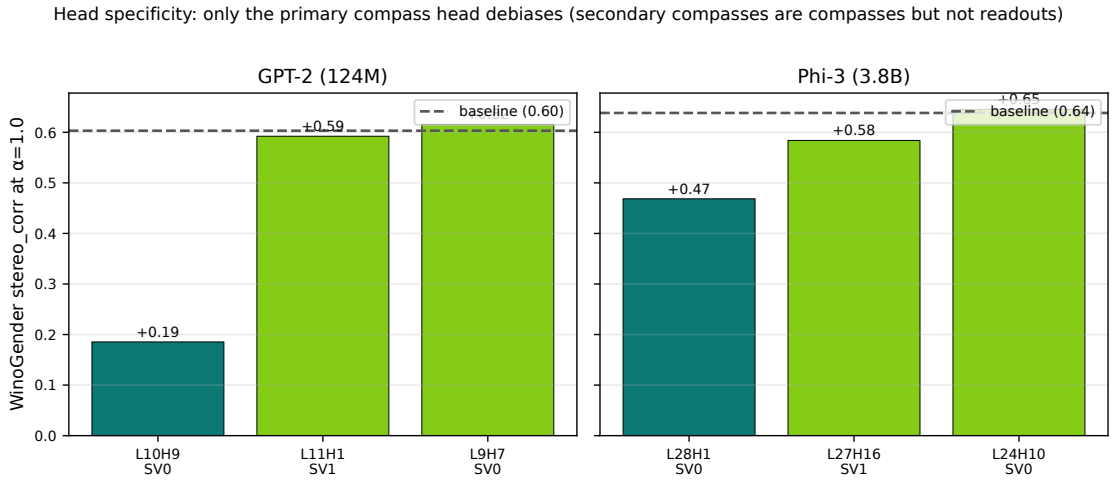

Figure 3: **Head specificity.** At α=1.0, only the *primary* compass head (dark teal) actually debiases; other passing compass heads in the same model (light green) leave `stereo_corr` near baseline. The dissociation rules out cherry-picking and supports a specific functional role for L10 H9 / L28 H1.

Finding

This is a *dissociation*: secondary compass heads carry pronoun-salience but *not* the occupation-to-gender mapping. The primary head is specifically the *occupational-stereotype readout*. That makes the finding richer, not weaker.

9 Step 7 — cross-benchmark check (StereoSet)

To avoid a "one-metric fluke" objection, we ran the same intervention on **StereoSet** — a completely different benchmark that scores the full-sentence log-probability of *stereotype*, *anti-stereotype* and *unrelated* completions.

Technical: StereoSet scoring

For each context with three candidate fillers and gold-labels $\{0, 1, 2\}$, we compute sum-log-prob of each candidate sentence under the model (optionally with hooks installed) and count, per pair, `Stereotype Score (SS)` = % where stereotype > anti, and `LMS` = % where either meaningful option beats the unrelated one. `ICAT` combines both: $\text{LMS} \times \min(\text{SS}, 100 - \text{SS})/50$.

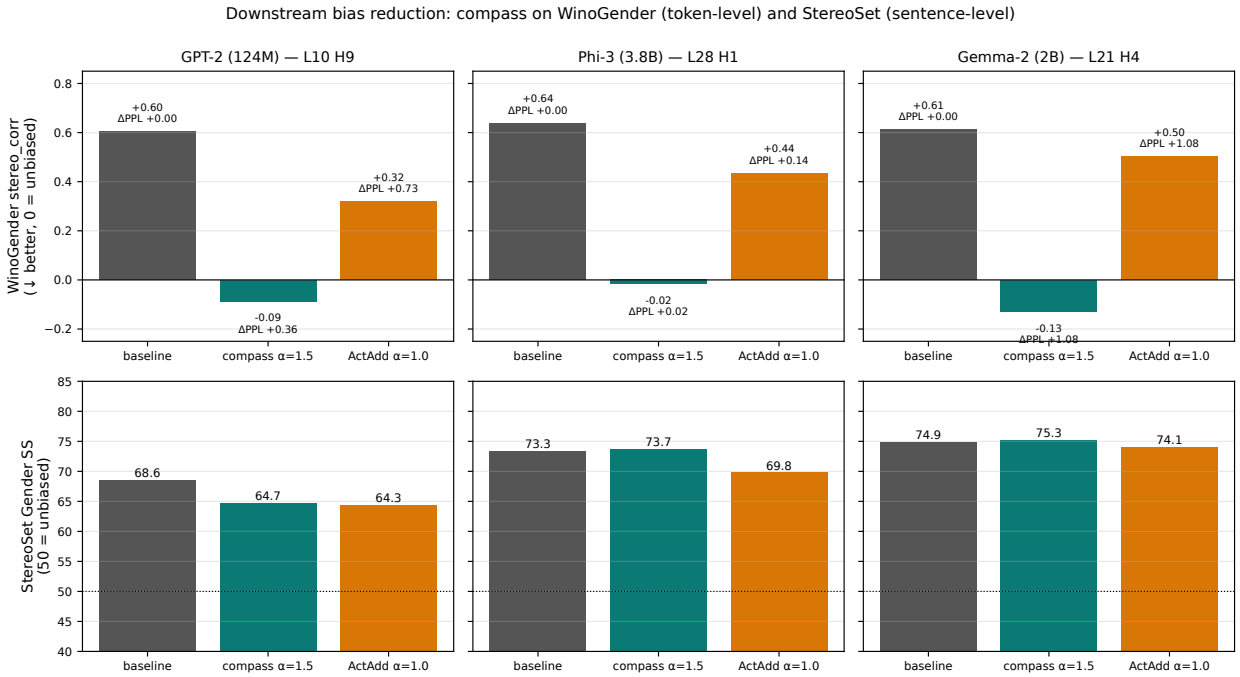

Figure 4: **Downstream bias reduction.** Top row: WinoGender `stereo_corr` (lower is better; zero is unbiased). Bottom row: StereoSet Gender SS (50 is unbiased). Compass works on all three models for WinoGender (token-level), but only GPT-2 for StereoSet (sentence-level).

Finding

WinoGender result replicates on all three models. StereoSet replicates on GPT-2 only; on Phi-3/Gemma the compass acts sharply at the pronoun slot but does not propagate into whole-sentence log-prob averages. That is **an honest boundary of the finding**: compass is a position-local, gender-specific direction, not a universal knob.

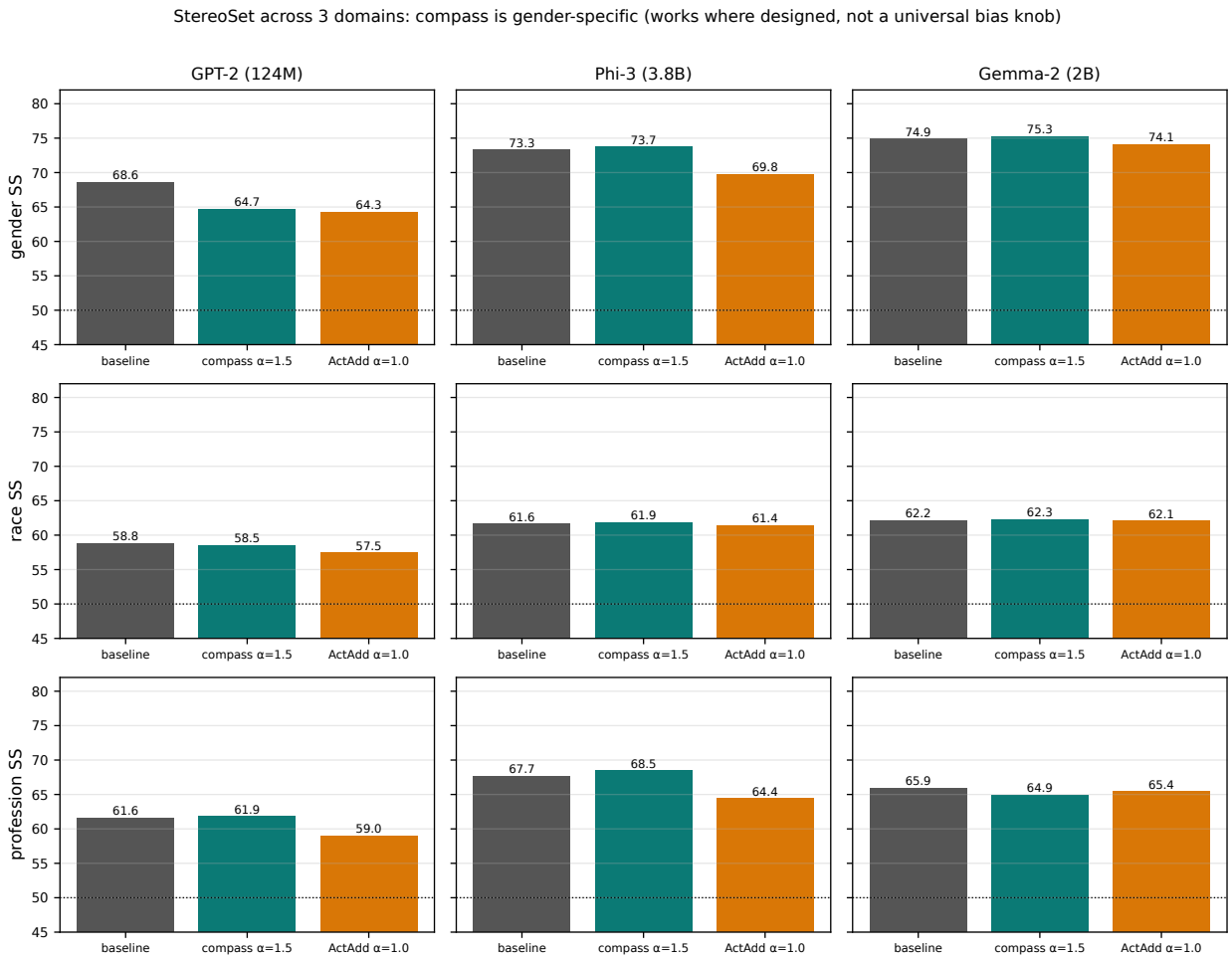

Figure 5: **StereoSet across three bias domains.** Compass barely moves race or profession SS; the *gender* compass is domain-specific and does not act as a universal bias knob. This is scientific evidence for specificity, not a bug.